Notes: Fecht, Hackethal, and Karabulut (JF, 2018)

Is Proprietary Trading Detrimental to Retail Investors?

Contents

1 Big picture

- **Question.** Does proprietary trading by universal banks harm their own retail customers? More specifically: when a bank sells a stock from its proprietary portfolio, do its retail customers buy the same stock, do those stocks subsequently underperform, and do customers of banks with proprietary trading desks earn lower portfolio returns?
- **Setting.** The paper studies German universal banks from **December 2005 to September 2009**. The key data source is the Deutsche Bundesbank's Security Deposits Statistics, which report security-by-security holdings for each German bank and for the aggregate retail customers of that bank.
- **Core conflict of interest.** A universal bank may have two roles under the same roof. Its proprietary desk wants to trade profitably and avoid market impact. Its retail arm advises or manages investments for clients. The conflict arises because the bank may have an incentive to place stocks it wants to sell into the portfolios of less informed retail customers.
- **Why this is an important setting.** Retail investors often rely on bank advice because financial advice is close to a *credence good*: customers may not be able to evaluate the quality of advice either before or after receiving it. Universal banks therefore may have both the information advantage and the distribution channel needed to shift unwanted positions to clients.
- **Main empirical idea.** The paper cannot observe direct trades between a bank and its customers. Instead, it compares quarterly changes in the bank's holding of stock i with quarterly changes in the aggregate holding of the bank's retail customers in the same stock.
- **Main baseline result.** When a bank *buys* a stock, customer flows in that stock are positively related to bank flows. But when a bank *sells* a stock, customer flows move in the opposite direction: the bank's retail customers tend to buy the stock the bank is selling.
- **Main causal result.** The paper uses the 2008 German rescue program, SoFFin, as a shock. Banks receiving state support were required to shut down most or all own-account trading and/or reduce risky asset holdings. When these forced sales occur, the same negative bank-customer flow relation appears, supporting a causal interpretation.
- **Main channel result.** The effect is stronger for banks with active portfolio management units. This suggests that discretionary portfolio management is one important channel through which stocks can move from the bank's trading book into customer portfolios.
- **Main motive result.** The effect is stronger for illiquid stocks and for large bank sell trades. This supports the price-impact motive: banks appear more likely to use customers as buyers when selling in the market would be more costly.
- **Main performance result.** Retail customers lose money on transactions in which they buy stocks that their bank sells. Stocks sold by banks to customers underperform customers' other holdings and other purchases.
- **Main portfolio-level result.** Retail customers of banks with proprietary trading desks underperform customers of banks without proprietary trading desks. The difference in four-factor alpha is about **17 basis points per month**, or about **2.04% per year**.

- **Economic takeaway.** Combining proprietary trading and retail banking can create a serious internal conflict of interest. Any informational benefits from a bank's trading expertise appear to be outweighed by the cost of being used as a buyer when the bank wants to unload positions.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: universal banks and retail advice

- German banks commonly combine retail banking, advisory services, portfolio management, and proprietary trading.
- Banks may have superior information and trading ability because they have better technology, human capital, market access, and firm relationships.
- The same structure can create agency problems because retail customers depend on the bank for advice and may not observe whether advice is independent or motivated by the bank's own trading needs.
- The paper focuses on single-stock holdings rather than bonds or mutual funds because individual stocks are more information-sensitive and more vulnerable to information asymmetry.

Interpretation. The paper asks whether the theoretical benefit of universal banking - better information and scope economies - is dominated by a conflict of interest between the trading desk and the retail client business.

2.2 Bundesbank security-holdings data

- The original data set contains security-by-security portfolio holdings for each German monetary financial institution and for the aggregate retail customers of each institution.
- The sample period is **December 2005 to September 2009**, with holdings observed at the end of each quarter.
- Retail customers include mass retail, private banking, and wealth management customers, but exclude institutional clients.
- Bank holdings include proprietary trading positions, market-making inventory, and strategic stock investments.
- Because strategic holdings should not change frequently, quarterly changes in bank stock holdings are interpreted as mainly reflecting changes in the bank's trading book.

Interpretation. The data are unusually broad because they cover the universe of German banks, but the empirical object is still a quarterly net-holdings change rather than an observed transaction between a bank and an individual customer.

2.3 Sample construction

- As of December 2005, **2,037** German monetary financial institutions reported holdings to the Bundesbank.
- Many small regional banks have very limited proprietary stock holdings. The main sample therefore keeps banks in the top **10th percentile** by the time-series average of quarterly stock portfolio value.
- The paper also requires the bank to have a retail banking unit.
- The final main sample contains **102 banks**, **8,203 different stocks**, and **133,177 bank-stock-quarter observations**.
- These observations represent about **63%** of stock investments made by all German monetary financial institutions during the period.

Interpretation. The main sample is designed to focus on banks for which proprietary stock trading is economically meaningful while still preserving a broad coverage of the German banking sector.

2.4 Portfolio composition of sample banks

- The average proprietary portfolio size of sample banks is about **8.24 billion euros**, compared with **1.61 billion euros** for all banks.
- Listed equities are about **5.8%** of sample-bank proprietary portfolios, compared with about **1.0%** for the average German monetary financial institution.
- The average value of listed equities held by a sample bank is about **481 million euros**.

Interpretation. The sample banks are not representative of every tiny local bank; they are the banks for which the proprietary equity-trading conflict is most relevant.

2.5 Stock holdings and stock characteristics

- The average bank holding in a given stock is about **2.51 million euros**.
- The corresponding average holding of that bank's aggregated retail customers in the same stock is about **2.20 million euros**.
- The mean stock portfolio of a bank is about **481 million euros**; the mean aggregate customer stock portfolio at that bank is about **833 million euros**.
- About **48%** of bank stock holdings have positive average returns in the previous quarter.
- Industrials and financials are the two largest sector groups in the stock sample, each accounting for roughly **17–18%** of stocks in the Internet Appendix decomposition.

Interpretation. The bank and retail customer positions are economically large enough that cross-holdings can reveal meaningful patterns in stock flows.

2.6 Normalized holdings and flow variables

For bank j, stock i, and quarter t, the paper defines normalized holdings as the fraction of the stock's free-float market capitalization held by the bank or by the bank's retail customers:

$$Share^{B}_{ijt} = \frac{Holdings^{B}_{ijt}}{FFMC_{it}}, \qquad Share^{C}_{ijt} = \frac{Holdings^{C}_{ijt}}{FFMC_{it}}. \tag{1}$$

The main flow variables are changes in these normalized holdings:

$$\Delta Share^{B}_{ijt} = \frac{Holdings^{B}_{ijt}}{FFMC_{it}} - \frac{Holdings^{B}_{ij,t-1}}{FFMC_{i,t-1}}, \tag{2}$$

$$\Delta Share^{C}_{ijt} = \frac{Holdings^{C}_{ijt}}{FFMC_{it}} - \frac{Holdings^{C}_{ij,t-1}}{FFMC_{i,t-1}}. \tag{3}$$

The paper converts these changes into basis points and trims the data at the **2.5%** level.

Interpretation. Normalizing by free-float market capitalization removes valuation effects, accounts for stock splits, and makes holdings comparable across small and large stocks.

2.7 Main sell indicator

The key sell dummy is

$$Sell^{B}_{ijt} = \begin{cases} 1, & \text{if } \Delta Share^{B}_{ijt} < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Interpretation. The main test is whether $\Delta Share^{C}_{ijt}$ responds differently to $\Delta Share^{B}_{ijt}$ when the bank is selling versus when the bank is buying.

2.8 Control variables

The baseline regressions include lagged stock-level controls:

- $DummyGain_{i,t-1}$: indicator for a positive average return in the previous quarter.
- $ReturnVolatility_{i,t-1}$: standard deviation of daily returns in the previous quarter.
- $MtBV_{i,t-1}$: market-to-book ratio.
- $TradingVolume_{i,t-1}$: average number of shares traded in the previous quarter.

Interpretation. These controls absorb basic return, risk, valuation, and liquidity conditions that may affect both banks' and customers' stock demand.

2.9 Univariate flow patterns

- In the full sample, the correlation between bank flows and customer flows in the same stock is positive: about **0.105**.
- Conditional on bank *sell* trades, the correlation is negative: about **-0.050**.
- Conditional on bank *buy* trades, the correlation is positive: about **0.106**.

Interpretation. The raw data already show an asymmetric pattern: customers move with the bank when the bank buys, but appear to move against the bank when the bank sells.

2.10 Customer sophistication comparison

- A possible concern is selection: more sophisticated customers might avoid banks with proprietary trading desks.
- The paper uses survey data on customer education to compare clients of banks with and without proprietary trading.
- Mean years of schooling are almost identical: about **13.21 years** in both groups, with a median of **11 years**.
- The similarity holds year by year from 2005 to 2011.

Interpretation. The portfolio-performance comparison is less likely to be driven simply by more financially literate households sorting away from banks with proprietary trading.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: why banks might sell to customers

- If a bank has private negative information about a stock, selling openly into the market can reveal information and create adverse price impact.
- Even if the bank sells for liquidity reasons, large sales can be front-run or interpreted as negative information by other traders.
- Retail customers can serve as a less price-sensitive buyer base because they rely on bank advice, analysts, relationship managers, discretionary portfolio managers, or affiliated funds.
- The price-impact motive predicts stronger effects when stocks are illiquid and when the bank's sell order is large.

Interpretation. The core mechanism is not that banks always have private information. The broader point is that selling to customers can help a bank avoid the trading costs associated with selling in the open market.

3.2 Baseline bank-customer flow regression

The main regression is

$$\Delta Share_{ijt}^{C} = \beta_1 \Delta Share_{ijt}^{B} + \beta_2 Sell_{ijt}^{B} + \beta_3 \left(\Delta Share_{ijt}^{B} \times Sell_{ijt}^{B}\right) + \beta_4 Controls_{i,t-1} + \gamma_j + \gamma_t + \gamma_k + \varepsilon_{ijt}. \tag{5}$$

Here:

- γ_j are bank fixed effects.
- γ_t are quarter fixed effects.
- γ_k are industry fixed effects.
- Standard errors are clustered at the bank level.

Interpretation. When $Sell_{ijt}^{B} = 0$, the slope between customer flows and bank flows is β_1. When $Sell_{ijt}^{B} = 1$, the slope is $\beta_1 + \beta_3$. The paper's central prediction is $\beta_1 + \beta_3 < 0$, meaning customer buying rises when the bank sells more.

3.3 Herding control

A concern is that customers buy the same stocks because all retail investors herd into a stock, not because their own bank is pushing that stock. The paper constructs a stock-level retail herding variable:

$$\Delta Share_{ijt}^{C,others} = \sum_{h \neq j} \Delta Share_{iht}^{C}. \tag{6}$$

Interpretation. This variable captures aggregate retail demand for stock i by customers of all other banks. If the main result survives this control, it is less likely to be a simple retail herding pattern.

3.4 Market-making and customer-demand checks

- The paper excludes stocks for which a bank is explicitly listed as a Deutsche Boerse designated sponsor.
- It controls for the lagged normalized customer holding in a stock, because customers may be less likely to buy more when they already hold a large position.
- It also estimates models with bank-time and stock-time fixed effects.

Interpretation. These checks address the alternative view that banks are merely acting as market-makers or passively accommodating customer demand.

3.5 Dynamic panel specification

Because banks may split large orders over several quarters, bank flows can be serially correlated. The paper therefore estimates dynamic panel models using Arellano-Bover GMM methods and bank-stock fixed effects.

Interpretation. This allows for persistent trading strategies and unobserved heterogeneity in the bank-stock relationship. The main negative sell-side relation still survives.

3.6 Forced-sales regression around SoFFin rescues

The causal test uses banks that received support from the German Special Fund for Financial Market Stabilization after the 2008 crisis. Beneficiary banks were required to shut down most or all own-account trading and/or materially reduce risky assets.

The forced-sales regression is

$$\Delta Share^{C}_{ijt} = b_1 \Delta Share^{B}_{ijt} + b_2 ForcedSales_{jt} + b_3 \left(\Delta Share^{B}_{ijt} \times ForcedSales_{jt}\right) + b_4 Controls_{i,t-1} + \gamma_j + \gamma_t + \gamma_k + \varepsilon_{ijt}. \tag{7}$$

Interpretation. If the same negative bank-customer relation appears when banks are forced to sell for regulatory reasons, then the evidence is closer to causal. The customer buying pattern is not merely a response to customers initiating trades first.

3.7 Portfolio-management channel

The paper splits banks according to whether they have an active portfolio management unit, and also estimates a three-way interaction:

$$AM_j \times \Delta Share^{B}_{ijt} \times Sell^{B}_{ijt}. \tag{8}$$

Interpretation. If discretionary portfolio management is a channel, the sell-side push effect should be stronger for banks with such units. The paper finds exactly that.

3.8 Illiquidity and price-impact tests

The paper measures stock illiquidity using the Amihud ratio:

$$Illiquidity_{it} = \frac{1}{D_{it}} \sum_{d=1}^{D_{it}} \frac{|r_{idt}|}{Vol_{idt}}, \tag{9}$$

where D_{it} is the number of trading days, r_{idt} is the daily return, and Vol_{idt} is daily trading volume.

It then interacts illiquidity indicators with the bank sell-flow interaction:

$$Illiquid_{it} \times \Delta Share^{B}_{ijt} \times Sell^{B}_{ijt}. \tag{10}$$

Interpretation. If banks sell to customers to avoid market impact, the pattern should be stronger when market impact is most severe: illiquid stocks, large bank sell trades, and crisis-period market illiquidity.

3.9 Trading-profit regression

To see whether customers lose when they buy what the bank sells, the paper estimates

$$Profit^{C}_{ijt} = b_1 Profit^{B}_{ijt} + b_2 Sell2C_{ijt} + b_3 \left(Profit^{B}_{ijt} \times Sell2C_{ijt}\right) + \gamma_j + \gamma_t + \varepsilon_{ijt}. \tag{11}$$

- $Profit^{B}_{ijt}$ is the bank's quarterly trading profit in stock i.
- $Profit^{C}_{ijt}$ is the corresponding trading profit of the bank's retail customers.
- $Sell2C_{ijt} = 1$ if the bank sells stock i and its retail customers buy stock i in the same quarter.

Interpretation. A negative coefficient on $Sell2C$ means customers lose on the transactions in which they buy what the bank sells. A negative interaction with bank profits means customer losses are larger when bank profits on the same stock are larger.

3.10 Portfolio-return comparisons

For each bank, the paper forms a *case group* and several control portfolios.

The case group contains stocks that banks sell while their retail customers buy. The control groups include:

- customers' other stock holdings;
- customers' other purchases;
- banks' other holdings;
- banks' purchases;
- banks' other sales not bought by customers.

Equal-weighted portfolio returns are

$$EWR^{G}_{jt} = \frac{1}{N} \sum_{i \in G_j} r_{it}, \tag{12}$$

where G indexes the case or control group.

Value-weighted portfolio returns are

$$VWR^{G}_{jt} = \sum_{i \in G_j} w^{G}_{ijt} r_{it}. \tag{13}$$

Interpretation. These comparisons ask whether the stocks transferred to customers are worse than what customers otherwise hold or buy, and worse than what banks otherwise hold or buy.

3.11 Full customer-portfolio comparison

The paper also compares the overall stock portfolio performance of customers of banks with and without proprietary trading desks. Performance measures include:

- raw monthly returns;
- one-factor alpha;
- four-factor alpha using Fama-French factors and momentum;
- characteristics-adjusted excess returns based on size and market-to-book benchmark portfolios.

Interpretation. This is the net-effect test. Even if customers lose on some pushed stocks, proprietary trading could still benefit customers through better information. The portfolio-level comparison asks whether the total effect is positive or negative.

4 Identification and empirical design (what makes the estimates credible)

4.1 The main identification problem

- The paper does not observe the exact counterparty to a bank's trade.
- It observes only quarterly net holdings by the bank and by the bank's aggregate retail customers.
- Therefore, a negative bank-customer flow relation could in principle reflect customers demanding a stock and the bank passively supplying it, rather than the bank actively pushing it.

Interpretation. The empirical challenge is to distinguish bank-driven sales to customers from customer-driven trades or broad retail demand shocks.

4.2 Asymmetry between bank buys and bank sells

- The strongest reduced-form evidence is the asymmetry.
- If banks were simply market-makers for customer demand, a negative relation might appear for both bank buys and bank sells.
- Instead, the negative relation appears specifically when banks sell stocks.
- When banks buy stocks, bank flows and customer flows are positively related.

Interpretation. The asymmetry is consistent with the idea that banks can place stocks into customer portfolios when they want to sell, but cannot similarly buy stocks from customers because retail customers generally face short-sale constraints and do not necessarily hold the stocks banks want to buy.

4.3 Controls for herding and retail demand

- The herding variable captures whether retail investors at other banks are also buying the same stock.
- The main negative sell-side relation survives after controlling for this aggregate retail flow.
- The paper also controls for customers' prior holdings to address pre-existing customer demand or concentration in a particular stock.

Interpretation. The result is not simply that all retail investors chase the same stock and banks happen to sell into that retail demand.

4.4 Controls for market-making

- A bank could sell to customers because it is acting as a market-maker, not because of a conflict of interest.
- The paper uses Deutsche Boerse designated-sponsor information to identify explicit bank-stock market-making relationships.
- Excluding these market-making stocks does not change the main result.

Interpretation. The result is not just normal liquidity provision by banks in their designated market-making roles.

4.5 Forced-sales design

- SoFFin support imposed regulatory restrictions on beneficiary banks' own-account trading and risky asset holdings.
- The paper treats these restrictions as an exogenous negative shock to banks' proprietary stock positions.
- Beneficiary sample banks reduced stock portfolios by about **46.1%**, or about **2.82 billion euros**, during the treatment period.
- The same bank-customer pattern appears for these forced sales.

Interpretation. This is the paper's most direct attempt to pin down direction of causality: when the bank is forced to sell, customers of that same bank buy the stocks.

4.6 Price-impact heterogeneity

- If the motive is to avoid price impact, the effect should be stronger for illiquid stocks.
- It should also be stronger for large bank sell trades.
- It should be stronger when market liquidity is poor, such as after the Lehman Brothers collapse.

- The paper finds evidence consistent with all three predictions.

Interpretation. The heterogeneity results strengthen the mechanism. They are not merely robustness checks; they show that the estimated behavior intensifies exactly where the theory predicts the bank's incentive is strongest.

4.7 Performance evidence as an indirect validation

- Stocks sold by banks to customers subsequently underperform customers' other holdings and purchases.
- Customers lose on the relevant transactions.
- Overall customers of banks with proprietary trading desks underperform customers of banks without proprietary trading desks.

Interpretation. If banks were simply sharing superior information with customers, the performance evidence should look different. The observed underperformance supports the conflict-of-interest interpretation.

4.8 Remaining limitations

- The paper cannot observe individual advice, individual investor decisions, or the exact trade counterparty.
- The quarterly frequency prevents very high-frequency timing tests.
- Bank holdings mix proprietary trading, market-making, and strategic positions, although quarterly changes are argued to mostly reflect trading-book changes.
- The paper identifies a strong institutional pattern, but it cannot fully distinguish whether the internal channel is analyst recommendations, relationship managers, discretionary portfolio management, or affiliated funds in every case.

Interpretation. The evidence is very strong for the existence of a bank-customer conflict, but the exact internal organizational pathway is only partially identified.

5 Tables and results

5.1 Table I: Portfolio composition

Read:

- Sample banks have much larger proprietary portfolios than the average German monetary financial institution: **8.24 billion euros** versus **1.61 billion euros**.

- Listed equities are about **5.8%** of sample-bank proprietary portfolios, compared with **1.0%** for all banks.
- The average listed-equity position in sample banks is about **481 million euros**.

Economic message. The paper studies banks for which proprietary equity trading is material. The sample is therefore appropriate for testing conflicts between equity trading desks and retail customers.

Table I

Portfolio Composition: Banks in the Final Sample versus All Banks in Germany

This table provides summary statistics for the portfolio compositions of banks in the final sample and those of all banks in Germany. Panel A describes portfolio shares of each asset class in the portfolios of sample banks and those of all banks in Germany. For comparison, Panel B presents the euro values of each asset class in the bank portfolios. The summary statistics are (1) the number of bank-quarter observations in each group, (2) the sample average of the portfolio share (value) of a given asset, and (3) the sample standard deviation of the portfolio share (value) of a given asset in each group. For variable definitions, see Table XIII. The data come from the Deutsche Bundesbank and cover the period from December 2005 to September 2009.

	All Banks			Banks in the Final Sample		
	No of Obs	Mean	*SD*	No of Obs	Mean	*SD*
Panel A: Portfolio Shares						
Portfolio size (in euro)	30,581	1.61e + 09	1.17e + 11	1,600	8.24e + 09	1.92e + 10
Short-term bonds share	30,581	0.0192	0.058	1,600	0.027	0.061
Long-term bonds share	30,581	0.790	0.235	1,600	0.714	0.198
Listed equities share	30,581	0.01	0.053	1,600	0.058	0.135
Nonlisted share	30,581	0.0133	0.085	1,600	0.003	0.010
Other equities share	30,581	0.001	0.01	1,600	0.0011	0.007
Mutual funds share	30,581	0.166	0.220	1,600	0.197	0.181
Panel B: Euro Values						
Short-term bonds	30,581	3.91e + 07	9.37e + 08	1,600	4.01e + 08	2.02e + 09
Long-term bonds	30,581	7.70e + 08	4.84e + 09	1,600	6.63e + 09	1.58e + 10
Listed equities	30,581	3.94e + 07	6.72e + 08	1,600	4.81e + 08	2.26e + 09
Nonlisted equities	30,581	3,508,774	7.05e + 07	1,600	5.34e + 07	2.98e + 08
Other equities	30,581	6.69e + 08	1.17e + 11	1,600	3,547,663	2.12e + 07
Mutual funds	30,581	9.29e + 07	5.37e + 08	1,600	6.71e + 08	1.60e + 09

Table II

Descriptive Statistics for the Final Sample

This table provides summary statistics for the dependent variables and control variables employed in the analysis. The summary statistics are (1) the number of observations for each variable, (2) the sample mean, (3) the sample standard deviation, (4) the 25th percentile of the distribution, (5) the sample median, and (6) the 75th percentile of the distribution for each variable. The summary statistics are computed across bank-stock observations and over time. Note that we winsorize all stock controls by setting the extreme values that fall above (below) the 99th (first) percentile to the corresponding upper (lower) boundary of the sample. For variable definitions, see Table XIII. The data come from the Deutsche Bundesbank and cover the period from December 2005 to September 2009.

	No of Obs	Mean	*SD*	25% percentile	Median	75% percentile
$Holdings^B$	133,177	2,513,902	2.10e + 07	4,879.33	91,609.56	543,380
$Holdings^C$	133,177	2,202,146	1.82e + 07	0	10,465.51	309,195.9
$\Delta Share^B$	133,177	0.542	3.785	−0.0175	0.003	0.289
$\Delta Share^C$	133,177	0.361	2.407	−0.0005	0	0.01
Dummy Gain	133,177	0.483	0.499	0	0	1
Return Volatility	133,177	0.033	0.029	0.0162	0.024	0.037
FFMC	133,177	1.19e + 10	2.56e + 10	3.17e + 08	2.10e + 09	1.02e + 10
Trading Volume(/100)	133,177	56.144	142.091	0.295	7.058	41.922
MtBV	133,177	2.396	2.754	1.08	1.771	2.903

5.2 Table II: Descriptive statistics

Read:

- The average holding of a bank in a given stock is about **2.51 million euros**; the average aggregate customer holding at the same bank is about **2.20 million euros**.
- The average quarterly change in normalized bank holdings is about **0.542 basis points**; for customer holdings it is about **0.361 basis points**.
- The stock sample contains a wide range of free-float market capitalizations, trading volumes, and market-to-book ratios.

Economic message. The holdings data are large enough to measure meaningful bank-customer co-movement, while normalization makes flows comparable across stocks.

5.3 Table III: Univariate bank-customer flow correlations

Read:

- Full-sample bank-customer flow correlation: **0.105**.
- Correlation when banks sell: **-0.050**.

- Correlation when banks buy: **0.106**.

Economic message. Before any regression controls, the key asymmetry is already visible: customer flows oppose bank flows when banks sell, but move with bank flows when banks buy.

Table III

Stock Investments of Banks and Their Retail Customers: Univariate Analysis

This table presents Pearson correlation coefficients for changes in the normalized stock holdings (i.e., $Holdings^B/FFMC$) of a bank and those of its retail customers in a given quarter. Column (1) reports the unconditional correlation between $\Delta Share^C_{ijt}$ and $\Delta Share^B_{ijt}$. Column (2) is restricted to the sell trades of banks, and column (3) considers only the buy trades of banks. The corresponding p-values of the pairwise correlations are reported in parentheses. The data come from the Deutsche Bundesbank and cover the period from December 2005 to September 2009.

	Full Sample $\Delta Share^C_{ijt}$ (1)	Sell Trades $\Delta Share^C_{ijt}$ (2)	Buy Trades $\Delta Share^C_{ijt}$ (3)
$\Delta Share^B_{ijt}$	0.105	−0.05	0.106
	(0.000)	(0.000)	(0.000)
No of Obs	133,177	53,200	79,642

Table IV

Comparison of Bank Customers by the Presence of a Proprietary Trading Unit

This table presents the mean and median years of schooling for customers of banks grouped by whether the bank has a proprietary trading desk. Information on educational attainment comes from an annual representative household panel survey conducted by a large German market research institute. See the Internet Appendix for further details about the household survey. The sample period covers the years between 2005 and 2011.

Years of Schooling	Banks without Proprietary Trading: No of Obs	Mean	*SD*	Median	Banks with Proprietary Trading: No of Obs	Mean	*SD*	Median	*t*-Test
Full sample (2005 to 2011)	1,564	13.205	3.018	11.00	2,976	13.210	3.021	11.00	−0.0541
2005	318	12.89	3.00	11.00	565	13.16	3.07	11.00	−1.2476
2006	264	13.44	3.06	11.00	487	13.42	3.07	11.00	0.0700
2007	219	13.32	3.06	11.00	339	13.14	3.01	11.00	0.6837
2008	176	13.39	3.04	11.00	339	13.14	2.99	11.00	0.8864
2009	176	13.24	3.06	11.00	328	13.22	3.02	11.00	0.0767
2010	181	12.78	2.84	11.00	367	13.04	2.94	11.00	−1.0025
2011	230	13.42	3.01	11.00	551	13.26	3.02	11.00	0.6664

5.4 Table IV: Customer sophistication comparison

Read:

- Customers of banks without proprietary trading average about **13.205** years of schooling.
- Customers of banks with proprietary trading average about **13.210** years of schooling.
- Median schooling is **11 years** in both groups.

Economic message. The paper's customer-performance comparison is less likely to be explained by less educated customers selecting into banks with proprietary trading.

5.5 Table V: Baseline flow regressions

Table V

Stock Investments of Banks and Their Retail Customers: Baseline Analysis

This table reports coefficient estimates from the baseline regressions. In column (1), we estimate the baseline regression. In column (2), we include in the model an additional control variable that measures the aggregate investment decisions of retail investors in a specific stock. In column (3), we remove those stock observations for which banks serve as market-makers. In column (4), we include as a control variable the lagged normalized holdings of a given stock in the portfolios of bank customers. In column (5), we include the lagged dependent variable in the estimation model. In columns (1) to (4), we estimate pooled panel regressions with bank, time, and industry fixed effects using ordinary least squares, while in column (5) we use generalized method of moments (GMM) estimators for dynamic models of panel data developed by Arellano and Bover (1995), controlling for bank-stock fixed effects. t-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. For variable definitions, see Table XIII. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $\Delta Share^C_{ijt}$ (1)	(2)	(3)	(4)	(5)
$\Delta Share^B_{ijt}$ (β_1)	0.058**	0.055**	0.058**	0.057**	0.0227***
	(3.29)	(3.17)	(3.09)	(3.33)	(5.28)
$Sell^B_{ijt}$ (β_2)	−0.415***	−0.379***	−0.400***	−0.414***	−0.123***
	(−6.11)	(−5.96)	(−6.23)	(−6.12)	(−11.88)
$\Delta Share^B_{ijt} \cdot Sell^B_{ijt}$ (β_3)	−0.076**	−0.062**	−0.066*	−0.067**	−0.038***
	(−2.66)	(−2.37)	(−2.55)	(−2.66)	(−5.50)
$Dummy\ Gain_{it-1}$	−0.106***	−0.055***	−0.100***	−0.102***	−0.028*
	(−4.72)	(−2.77)	(−4.79)	(−4.56)	(−2.55)
$Return\ Volatility_{it-1}$	2.334	2.596	2.314	2.315	1.178***
	(1.69)	(1.94)	(1.70)	(1.71)	(5.34)
$MtBV_{it-1}$	−0.011**	−0.008*	−0.010*	−0.008**	−0.0044*
	(−2.94)	(−2.40)	(−2.60)	(−2.64)	(−2.28)

(Continued)

Table V—*Continued*

	Regressand: $\Delta Share^C_{ijt}$ (1)	(2)	(3)	(4)	(5)
$Trading\ Volume_{it-1}/100$	−0.001***	−0.001***	−0.001***	−0.0006***	−0.002***
	(−5.60)	(−4.94)	(−5.47)	(−5.36)	(−8.57)
$\Delta Share^{others}_{it}$	–	0.035***	–	–	–
	–	(3.88)	–	–	–
$Share^C_{ijt-1}$	–	–	–	34.056***	–
	–	–	–	(7.60)	–
$\Delta Share^C_{ijt-1}$	–	–	–	–	−0.0192
	–	–	–	–	(−1.36)
					–
Constant	0.593***	0.320***	0.551***	0.574***	0.0896***
	(8.27)	(3.57)	(8.15)	(8.09)	(4.27)
Bank fixed effects	Yes	Yes	Yes	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	No
Bank-stock fixed effects	No	No	No	No	Yes
Clustering	Bank level	Bank level	Bank level	Bank level	Bank level
R^2	0.054	0.071	0.055	0.068	–
No of Obs	133,177	133,177	131,491	133,177	95,575
Hansen test (p-value)	–	–	–	–	0.422
Serial Correlation test (p-value)	–	–	–	–	0.308

Read:

- The coefficient on bank flow is positive when banks are not selling, consistent with banks and customers sometimes moving together.
- The interaction between bank flow and the sell dummy is negative and significant.

Table VI
Stock Investments of Banks and Their Retail Customers: Rescue Interventions of 2008 and Forced Sales

This table reports coefficient estimates from the regression specification shown in (2). In column (1), $ForcedSales^B_{jt}$ is an indicator variable that identifies those banks that received state aid and SoFFin guarantees during our observation period and thus were required to substantially reduce the volume of risky assets in their trading books at the beginning of the assistance period. In column (2), we also account for the adverse effects of the financial crisis period by introducing the indicator variable $Crisis_t$, which takes a value of one for the sample period following 2008Q3, and zero otherwise. t-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. For variable definitions, see Table XIII. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $\Delta Share^C_{ijt}$	
	(1)	(2)
$\Delta Share^B_{ijt}$ (b_1)	0.0615***	0.0627***
	(4.70)	(4.34)
$Forced\ Sales^B_{jt}$ (b_2)	−0.299**	−0.299**
	(2.69)	(2.69)
$\Delta Share^B_{ijt} \cdot Forced\ Sales^B_{jt}$ (b_3)	−0.0784***	−0.0752***
	(−5.58)	(−4.55)
$Dummy\ Gain_{it-1}$	−0.096***	−0.096***
	(4.30)	(4.30)
$Return\ Volatility_{it-1}$	2.515	2.519
	(1.83)	(1.37)
$MtBV_{it-1}$	−0.010**	−0.010**
	(−3.07)	(−3.06)
$Trading\ Volume_{it-1}/100$	−0.001***	−0.001***
	(−5.69)	(−5.69)
$Crisis_t$	–	−0.117
	–	(−1.15)
$\Delta Share^B_{ijt} \cdot Crisis_t$	–	−0.004
	–	(−0.31)
Constant	0.0465***	0.464***
	(5.99)	(6.06)
Bank fixed effects	Yes	Yes
Time fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Clustering	Bank level	Bank level
R^2	0.048	0.0481
No of Obs	133,177	133,177

- The implied sell-side slope, $\beta_1 + \beta_3$, is negative: when banks reduce a stock position, their retail customers tend to increase holdings in that stock.
- The result survives controls for retail herding, exclusion of market-making stocks, lagged customer holdings, and dynamic panel estimation.

Economic message. The baseline empirical fact is not generic co-movement between bank and customer portfolios. It is a one-sided sell-channel: customers buy when their bank sells.

5.6 Table VI: SoFFin forced-sales test

Read:

- The forced-sales interaction is negative and statistically significant.
- In the main specification, $\Delta Share^B \times ForcedSales$ is about **-0.0784**; after controlling for the crisis period, it is about **-0.0752**.
- The estimated effects are similar in magnitude to the baseline regressions.

Economic message. The causal evidence supports the paper's interpretation: when banks are forced to sell, their own retail customers buy the sold stocks.

Table VII

Stock Investments of Banks and Their Retail Customers: Additional Robustness and Sensitivity Checks

This table reports regression results for various robustness checks. In column (1), we use changes in the share of stocks scaled by the portfolio values of banks and retail customers. In column (2), we remove financial stocks from the sample. In column (3), we expand our sample to include all banks with proprietary trading units that have a quarterly average stock portfolio value of more than 5 million euros. In column (4), we only consider those stock observations for which banks hold less than 3% of the free-float market capitalization. In column (5), we consider only those stock observations in which both sample banks and their retail customers have positive holdings in their respective portfolios. For variable definitions, see Table XIII. In columns (1) to (5), we estimate pooled panel regressions with bank, time, and industry fixed effects using ordinary least squares; in columns (6) and (7), we include stock-time and bank-time fixed effects in our baseline model, respectively. *t*-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $\Delta Share^C_{ijt}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta Share^B_{ijt}$	0.0375***	0.0556**	0.0593**	0.0592**	0.1603***	0.0573**	0.0616**
	(9.47)	(3.25)	(3.31)	(3.28)	(5.08)	(3.19)	(3.23)
$Sell^B_{ijt}$	−0.3820***	−0.4422***	−0.4115***	−0.4137***	−1.1147***	−0.404***	−0.421***
	(−5.22)	(−6.03)	(−6.21)	(−6.08)	(−9.51)	(−6.01)	−0.075
$\Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	−0.0400***	−0.0769*	−0.0759**	−0.0775**	−0.1928***	−0.072*	(−1.97)
	(−5.22)	(−2.48)	(−2.67)	(−2.66)	(−4.14)	(−6.01)	–
$Dummy\ Gain_{it-1}$	0.3923***	−0.1020***	−0.1044**	−0.1035***	−0.225***	−0.104***	–
	(5.25)	(−4.15)	(−4.76)	(−4.62)	(−5.50)	(−4.64)	–
$Return\ Volatility_{it-1}$	−1.3907**	1.7996	2.350	2.358	2.038***	2.477	–
	(−3.15)	(1.43)	(1.73)	(1.70)	(3.54)	(1.75)	–

(Continued)

Table VII—*Continued*

	Regressand: $\Delta Share^C_{ijt}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$MtBV_{it-1}$	−0.0085***	−0.0131***	−0.011**	−0.0105**	−0.032***	−0.010**	–
	(−3.42)	(−3.47)	(−2.96)	(−2.93)	(−3.67)	(−2.76)	–
$Trading\ Volume_{it-1}/100$	−0.0002	−0.0008***	−0.0006***	−0.0007***	−0.0017***	−0.0007***	–
	(−0.75)	(−5.92)	(−5.60)	(−5.58)	(−6.35)	(−5.64)	–
Constant	0.6462***	0.6573***	0.589***	0.5889***	0.872***	0.497***	0.0461***
	(5.39)	(7.99)	(8.48)	(8.18)	(3.78)	(11.29)	(6.09)
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	No
Bank-time fixed effects	No	No	No	No	No	Yes	No
Stock-time fixed effects	No	No	No	No	No	No	Yes
Clustering	Bank level	Bank level	Bank level	Bank level	Bank level	Bank level	Bank level
R^2	0.112	0.055	0.055	0.055	0.058	0.074	0.395
No of Obs	139,864	107,472	135,929	132,813	85,415	133,177	133,177

5.7 Table VII: Robustness and sensitivity checks

Read:

- The result survives an alternative normalization by portfolio size rather than free-float market capitalization.
- It survives excluding financial stocks.
- It survives expanding the sample from **102** to **120** banks with proprietary trading units.
- It survives excluding large positions above the **3%** disclosure threshold.
- It survives restricting to bank-stock observations where both the bank and its customers have positive holdings.
- It also survives stock-time and bank-time fixed effects.

Economic message. The central pattern is not driven by financial-sector stocks, large block positions, the sample-selection cutoff, or time-varying shocks at the stock or bank level.

5.8 Table VIII: Role of portfolio management

Read:

- For banks with active portfolio management units, the sell interaction is negative and significant: about **-0.092**.
- For banks without active portfolio management units, the interaction is small and insignificant: about **-0.010**.
- In the full sample, the three-way interaction involving active portfolio management is negative and significant: about **-0.0848**.

Economic message. Discretionary portfolio management appears to be one important organizational channel through which banks can move stocks from their own trading books to customer portfolios.

Table VIII

Stock Investments of Banks and Their Retail Customers: Role of Portfolio Management

This table reports coefficient estimates for the analysis in which we estimate the baseline regressions for subsamples of banks divided according to whether they have an active portfolio management unit. In column (1), we estimate the baseline regression for banks that have an active portfolio management unit. In column (2), we estimate the baseline regression for banks that do not have an active portfolio management unit. In column (3), we estimate the baseline regression for the full sample using the three-way interaction between AM_{jt}, $\Delta Share^B_{ijt}$, and $Sell^B_{ijt}$. In specifications (1), (2), and (3), we estimate pooled panel regressions with bank, time, and industry fixed effects using ordinary least squares. t-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. For variable definitions, see Table XIII. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $\Delta Share^C_{ijt}$		
	Banks with AM (1)	Banks without AM (2)	Full Sample (3)
$\Delta Share^B_{ijt}$	0.0624*	0.046***	0.045***
	(2.68)	(3.60)	(0.011)
$Sell^B_{ijt}$	−0.442***	−0.370**	−0.357**
	(−5.28)	(−3.18)	(−2.94)
$\Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	−0.092*	−0.01	−0.038*
	(−2.41)	(−0.73)	(−2.62)
$Dummy\ Gain_{it-1}$	−0.113**	−0.079**	−0.105***
	(−3.59)	(−3.44)	(−4.74)
$Return\ Volatility_{it-1}$	4.639	0.918	2.356
	(1.79)	(0.77)	(1.72)
$MtBV_{it-1}$	−0.010*	−0.011	−0.011**
	(−2.12)	(−1.83)	(−2.84)
$Trading\ Volume_{it-1}/100$	−0.001***	−0.0004***	−0.0001***
	(−5.40)	(−2.83)	(−5.69)
$AM_{jt} \cdot \Delta Share^B_{ijt}$	–	–	0.0173
	–	–	(0.65)
$AM_{jt} \cdot Sell^B_{ijt}$	–	–	−0.084
	–	–	(−0.57)
$AM_{jt} \cdot \Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	–	–	−0.0848*
	–	–	(−2.04)
Constant	0.648***	0.397***	0.592***
	(6.51)	(4.34)	(8.30)
Bank fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Clustering	Bank level	Bank level	Bank level
R^2	0.046	0.094	0.054
No of Obs	90,491	42,686	133,177

Table IX

Stock Investments of Banks and Their Retail Customers: Price Impact Motive

This table reports coefficient estimates from regressions that test the price impact motive. In column (1), we include the three-way interaction term between the indicator variable characterizing illiquid stocks, the dummy variable for banks' sell trades, and changes in banks' normalized stock holdings. $Illiquid1_{it}$ takes the value of one if the Amihud ratio of stock i exceeds the sample median in quarter t. In column (2), we redefine illiquid stocks such that $Illiquid2_{it}$ takes the value of one if the Amihud ratio of stock i falls in the bottom tercile of stocks. In column (3), we restrict attention to relatively larger sell trades of banks, which we define as those that fall into the bottom first percentile of changes in banks' normalized stock holdings. In column (4), we again consider relatively larger sell trades of banks and account for bank-stock fixed effects. In specifications (1), (2), and (3), we estimate pooled panel regressions with bank, time, and industry fixed effects using ordinary least squares. In column (4), we estimate pooled panel regressions with bank-stock and time fixed effects using ordinary least squares. For variable definitions, see Table XIII. t-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $\Delta Share^C_{ijt}$			
	(1)	(2)	(3)	(4)
$\Delta Share^B_{ijt}$	0.010	0.006	0.0682***	0.0523***
	(1.49)	(0.63)	(4.31)	(3.87)
$Sell^B_{ijt}$	−0.166***	−0.1645***	−1.169**	−0.697
	(−4.52)	(−3.51)	(−3.09)	(−1.64)
$\Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	0.006	0.0304*	−0.1542***	−0.0895*
	(0.58)	(2.00)	(−3.55)	(−2.36)
$Dummy\ Gain_{it-1}$	−0.113***	−0.124***	−0.0957***	−0.0608**
	(−4.70)	(−3.97)	(−4.28)	(−3.20)
$Return\ Volatility_{it-1}$	0.716	−0.385	2.455	2.495*
	(0.63)	(−0.38)	(1.79)	(2.03)
$MtBV_{it-1}$	−0.006	−0.005	−0.010**	−0.003
	(−1.94)	(−1.25)	(−2.96)	(−0.87)
$Trading\ Volume_{it-1}/100$	−0.00001	0.0001	−0.0007***	−0.000001
	(−0.14)	(0.74)	(−5.69)	(0.29)
$Illiquid1_{it}$	0.6794***	–	–	–
	(5.49)	–	–	–
$Illiquid1_{it} \cdot Sell^B_{ijt}$	−0.521***	–	–	–
	(−4.08)	–	–	–
$Illiquid1_{it} \cdot \Delta Share^B_{ijt}$	0.065***	–	–	–
	(5.83)	–	–	–
$Illiquid1_{it} \cdot \Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	−0.1087***	–	–	–
	(−3.94)	–	–	–
$Illiquid2_{it}$	–	1.0598***	–	–
	–	(5.00)	–	–
$Illiquid2_{it} \cdot Sell^B_{ijt}$	–	−0.699***	–	–
	–	(−3.86)	–	–
$Illiquid2_{it} \cdot \Delta Share^B_{ijt}$	–	0.0754***	–	–
	–	(6.06)	–	–

(Continued)

Table IX—*Continued*

	Regressand: $\Delta Share^C_{ijt}$			
	(1)	(2)	(3)	(4)
$Illiquid2_{it} \cdot \Delta Share^B_{ijt} \cdot Sell^B_{ijt}$	–	−0.1401***	–	–
	–	(−4.16)	–	–
Constant	0.278***	0.214	0.452***	0.395***
	(3.57)	(1.70)	(6.07)	(6.06)
Bank fixed effects	Yes	Yes	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	No
Bank-stock fixed effects	No	No	No	Yes
Clustering	Bank level	Bank level	Bank level	Bank level
R^2	0.069	0.081	0.049	0.406
No of Obs	133,177	88,631	133,177	133,177

5.9 Table IX: Illiquidity, order size, and price impact

Read:

- The bank sell-customer buy relation is stronger for stocks classified as illiquid using the Amihud ratio.
- The effect remains when illiquidity is defined more conservatively by comparing the highest and lowest illiquidity terciles.
- The effect is also stronger for large bank sell trades.
- The large-trade result remains after including bank-stock fixed effects.

Economic message. The heterogeneity aligns with the price-impact mechanism. Banks are more likely to shift sales to customers when open-market selling would be especially costly.

5.10 Table X: Trading profits and losses

Read:

- The $Sell2C$ dummy is negative: retail customers lose on transactions in which they buy stocks that their bank sells.
- In the baseline specification, the loss associated with $Sell2C$ is about **13,146 euros**.
- The interaction between bank profit and $Sell2C$ is negative, around **-0.166**, meaning customer losses increase when the bank's trading profits in that stock increase.
- The result survives bank-time and bank-stock fixed effects.

Economic message. The flow relation has direct performance consequences. It is not just a harmless reshuffling of inventory; customers lose on these transactions.

Table X

Stock Performance: Trading Profits and Losses of Retail Investors

This table reports coefficient estimates from the trading profit/loss regressions as shown in equation (4). $Profit^B_{ijt}$ and $Profit^C_{ijt}$ represent the quarterly trading profits of bank j and its retail customers for a given transaction in stock i, respectively. $Sell2C^B_{ijt}$ is an indicator variable that takes the value of 1 if bank j sells stock i and its retail customers buy this stock in quarter t, and zero otherwise. In column (1), we estimate pooled panel regressions with bank and time fixed effects using ordinary least squares. In column (2), we estimate pooled panel regressions with bank-time fixed effects using ordinary least squares. In column (3), we estimate pooled panel regressions with bank-stock fixed effects using ordinary least squares. For variable definitions, see Table XIII. t-statistics, reported in parentheses, are calculated by correcting the standard errors by clustering at the bank level. ***, **, and * denote significance at the 0.001, 0.01, and 0.05 level, respectively. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Regressand: $Profit^C_{ijt}$		
	(1)	(2)	(3)
$Profit^B_{ijt}$	0.158***	0.154***	0.149***
	(5.99)	(6.00)	(5.34)
$Sell2C^B_{ijt}$	−13,145.81***	−12,548.81***	−6,273.26*
	(−4.24)	(−4.30)	(−1.82)
$Sell2C^B_{ijt} \cdot Profit^B_{ijt}$	−0.166***	−0.161***	−0.168***
	(−4.66)	(−4.67)	(−3.89)
Constant	55,570.76***	−3,284.94***	59,228.8***
	(4.86)	(−10.69)	(4.09)
Bank fixed effects	Yes	No	Yes
Bank-time fixed effects	No	Yes	No
Bank-stock fixed effects	No	No	Yes
Time fixed effects	Yes	No	No
Clustering	Bank level	Bank level	Bank level
R^2	0.11	0.156	0.313
No of Obs	132,269	132,269	132,269

Table XI

Stock Performance: Trading Profits and Losses of Retail Investors

This table presents monthly return comparisons of different stock portfolios for both equal and value weightings at the bank level. Note that the analysis excludes those months in which there was no stock in a given portfolio. We use the t-test (with unequal variances) and the Wilcoxon test to assess whether the mean and median returns of the stocks in the case and control groups are significantly different from one another at the bank level. The *Case Group* comprises stocks that are sold by banks to their retail customers. *Control Group I* includes all of the stock holdings of the retail customers of a bank other than those in the *Case Group*. *Control Group II* consists of the purchases of retail investors of a bank, excluding those stocks that their respective banks simultaneously sell from their proprietary portfolios. *Control Group III* includes all stocks held by a bank other than those stocks in the *Case Group*. *Control Group IV* comprises the purchases of banks. *Control Group V* comprises the sales of banks. To account for stock-specific characteristics, we use excess stock returns computed using characteristics-based benchmarks. Using the full universe of stocks that the sample banks and their retail investors hold in their portfolios (i.e., 16,165 different securities), we begin by constructing passive benchmark portfolios based on a double sort on each firm's market equity value and market-to-book ratio. On each formation date, we sort the sample stocks into quintiles based on each firm's end-of-June market value from the previous year. Next, we sort the stocks in each size quintile into further quintiles based on their market-to-book ratio, the end-of-December value from the previous year. This yields 25 benchmark portfolios in each year. To ensure that the performance comparisons are not driven by extreme return observations, we winsorize the monthly portfolio returns at the 1% level. The excess return of a given stock is computed as its monthly return in excess of that of the benchmark portfolio to which the stock belongs. The average monthly return for each benchmark portfolio aggregates over all the component stocks using value weighting in the portfolio. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	Equal-Weighted Portfolios					*Value-Weighted Portfolios*				
	No of Obs	Mean	t-Test	Median	Wilcoxon	No of Obs	Mean	t-Test	Median	Wilcoxon
Case Group vs.		−0.0033		0.0011			−0.0005		−0.0002	
Control Group I	2,490	−0.0008	−2.152	0.0009	−0.411	2,490	0.006	−5.399	−0.0004	−5.995
Control Group II	2,490	0.0206	−19.996	0.0220	−22.058	2,490	0.023	−17.288	−0.0011	−18.168
Control Group III	2,459	0.0002	−2.809	0.0021	−2.561	2,459	0.0033	−2.752	−0.0002	−2.463
Control Group IV	2,388	0.0088	−9.245	0.0082	−10.165	2,388	0.0119	−8.708	−0.0005	−8.368
Control Group V	2,285	−0.0104	4.825	−0.0068	7.861	2,285	−0.0079	4.727	0.0002	6.396

5.11 Table XI: Performance of stocks sold to customers

Read:

- The case-group stocks are those sold by banks and bought by their retail customers.
- These stocks underperform banks' other holdings and banks' purchases.
- The benchmark-adjusted return difference between bank purchases and the case group is about **1.21%**.
- The case group also underperforms customers' other stock holdings and customers' other stock purchases.
- The case group does not significantly underperform the bank's other sell trades, suggesting the case-group sales are not uniquely bad relative to other bank sales.

Economic message. Banks sell customers stocks that look worse than what customers otherwise hold or buy. This is consistent with banks using customers to absorb poor or costly-to-sell positions.

5.12 Table XII: Overall customer portfolio performance

Read:

- The paper expands the comparison to **1,863** banks with customer stock-holding data.
- Of these, **687** have no proprietary stock investments, while **1,176** have positive proprietary stock investments.
- Customers of banks without proprietary trading have higher performance across raw returns, one-factor alpha, four-factor alpha, and characteristics-adjusted excess returns.
- Mean four-factor alpha is about **-0.0072** for customers of banks without proprietary trading and about **-0.0089** for customers of banks with proprietary trading.
- The difference is about **17 basis points per month**, or roughly **2.04% annually**.

Economic message. The net effect of being at a bank with proprietary trading appears negative. Potential information benefits from bank trading expertise do not offset the conflict-of-interest cost.

5.13 Table XIII: Variable definitions

Read:

- The table collects the paper's key definitions: normalized holdings, changes in normalized holdings, sell and buy dummies, stock controls, herding measure, and illiquidity indicators.

Economic message. The empirical design is built on a simple object: how bank and customer holdings in the same stock change in the same quarter.

Table XII

Is Proprietary Trading Truly Detrimental to Retail Investors?

This table presents mean and median differences across the stock portfolios of retail customers of banks with proprietary trading divisions and the stock portfolios of customers of banks without proprietary trading units. We first calculate the monthly raw returns for each bank's customer portfolio that would be realized by holding the same portfolio share of stocks as reported by the end of the previous quarter in each month of the following quarter. We then compute equal-weighted monthly portfolio returns, which gives us 48 equal-weighted raw return observations for each bank's customer portfolio. Using the monthly returns, we calculate the monthly one-factor and four-factor alphas for each of these customer portfolios. The four-factor model includes the Fama and French (1993) factors and the Carhart (1997) momentum factor. We also use the time-series average of equal-weighted raw returns and excess stock returns computed using characteristics-based benchmarks. To ensure that the performance comparisons are not driven by extreme portfolio returns, we winsorize all performance measures at the 1% level. See the notes to Table XI for the definition of excess returns. The data come from the Deutsche Bundesbank and Thomson Reuters Datastream and cover the period from December 2005 to September 2009.

	No of Obs		Banks without Proprietary Trading	Banks with Proprietary Trading	t-Test/Wilcoxon Test
Raw Returns	687	Mean	−0.0088	−0.0105	7.590
	1,176	Median	−0.0100	−0.0109	8.945
One-Factor Alpha	687	Mean	−0.0091	−0.0106	7.423
	1,176	Median	−0.0101	−0.0109	8.687
Four-Factor Alpha	687	Mean	−0.0072	−0.0089	7.883
	1,176	Median	−0.0083	−0.0092	9.488
Excess Returns	687	Mean	0.0003	−0.00002	3.918
	1,176	Median	0.0001	−0.0001	3.517

Table XIII

Variable Definitions

Variable	Description
$Share^{C}_{ijt}$	Percentage of company shares i held by the retail customers of bank j in quarter t.
$Share^{B}_{ijt}$	Percentage of company shares i held by bank j in quarter t.
$Holdings^{B}_{ijt}$	Euro value of holdings of bank j in a given stock i in time t.
$Holdings^{C}_{ijt}$	Euro value of holdings of the retail customers of bank j in a given stock i in time t.
$FFMC_{it}$	Free-float market capitalization of stock i in quarter t.
$\Delta Share^{C}_{ijt}$	Changes in the normalized stock holdings of the retail customers of bank j in stock i in time t.
$\Delta Share^{B}_{ijt}$	Changes in the normalized stock holdings of bank j in stock i in time t.
$Sell^{B}_{ijt}$	A dummy variable equal to one if bank j sells stock i from its proprietary portfolio.
Buy^{B}_{ijt}	An indicator variable equal to one if bank j buys stock i in time t.
$Dummy\ Gain_{it-1}$	A dummy variable equal to one if stock i displayed positive average returns in the previous quarter, and zero otherwise.
$Return\ Volatility_{it-1}$	Standard deviation of the daily returns of stock i within quarter $t-1$.
$Trading\ Volume_{it-1}$	Quarterly average number of shares traded for stock i as measured in the previous quarter.
$MtBV_{it-1}$	Quarterly average of the market value of common equity divided by the balance sheet value of common equity in the company as measured in time $t-1$.
$\Delta Share^{others}_{it}$	Aggregate investment decisions of retail investors (other than bank j) in stock i at time t.
$Illiquid1_{it}$	An indicator variable that is equal to one if the Amihud ratio of stock i is above the median Amihud ratio across all stocks in quarter t, and zero otherwise.
$Illiquid2_{it}$	A dummy variable that is equal to one if the Amihud ratio of stock i is falls into the highest tercile based on the Amihud ratio across all stocks in quarter t, and zero otherwise.

6 Short summary

- The paper studies whether banks with proprietary trading desks exploit retail customers by selling them stocks from the bank's own portfolio.
- The setting is Germany, using security-level holdings of every bank and each bank's aggregate retail customers from December 2005 to September 2009.
- The baseline finding is asymmetric: bank buys move positively with customer flows, but bank sells are associated with customer purchases.
- The result survives many controls: herding, market-making exclusion, customer prior holdings, alternative normalizations, sample changes, fixed effects, and dynamic panel methods.
- The SoFFin forced-sales shock strengthens the causal interpretation.
- Portfolio management units appear to be an important channel.
- Illiquidity and large-order tests support the price-impact motive.
- Customers lose on transactions in which they buy stocks their bank sells.
- Stocks sold by banks to customers underperform customers' other holdings and purchases.
- Customers of banks with proprietary trading desks underperform customers of banks without proprietary trading desks.

7 Conclusion

7.1 Core conclusion

The paper concludes that proprietary trading can be detrimental to retail investors. The evidence shows that banks tend to sell some stocks from their proprietary portfolios to their own customers, especially when those stocks are illiquid or when the bank's sale is large.

7.2 Economic intuition

The economic intuition is straightforward. A bank wants to sell a stock without depressing the price or revealing negative information to the market. Retail customers are less informed and can be influenced through advice, portfolio management, or affiliated investment channels. This gives the bank a way to reduce market impact at the expense of customers.

7.3 Regulatory takeaway

The paper contributes to debates over universal banking, proprietary trading restrictions, and the separation of investment banking from retail banking. Although regulations such as the Volcker Rule are often motivated by systemic risk and moral hazard, this paper shows an additional investor-protection rationale: separating proprietary trading from retail distribution may reduce conflicts of interest that harm households.

7.4 Final takeaway

Universal banks may create information benefits, but the paper's evidence suggests that, in this context, the conflict of interest dominates. The retail customer receives not the full benefit of the bank's trading expertise, but some of the cost of the bank's need to unload positions.